Enterprise Automated ETL Data Pipeline & BI Platform

Complete Project Setup Guide for Any New Laptop / PC

Developer: Sami Siddiqui (@SamiSiddiqui45) | GitHub Project

GitHub Repo: https://github.com/SamiSiddiqui45/Enterprise-Automated-ETL-Data-Pipeline-BI-Analytics-Platform

Target Database: SQL Server (AnalyticsDW)

Overview: Yeh guide aapko step-by-step batati hai ke kisi dost ke laptop ya university ke PC par sirf apna GitHub repository clone karke SQL Server database, Python ETL pipeline, aur interactive BI analytics dashboard kaise 5 minute mein live run karna hai.

0. Prerequisites (Dost ke Laptop par yeh 3 cheezein hon)

• **Python 3.10 ya 3.12:** Installed with pip.
• **VS Code:** Code editing aur terminal chalane ke liye.
• **Microsoft SQL Server Express & SSMS:** Locally running database instance (.\SQLEXPRESS).

Step 1: Project Ko Laptop Mein Download Karein

Tareeqa A (Git se Clone karein - Recommended):

VS Code open karein, terminal (Ctrl + `) kholein aur yeh command run karein:

```
git clone https://github.com/SamiSiddiqui45/Enterprise-Automated-ETL-Data-Pipeline-BI-Analytics-Platform.git
cd Enterprise-Automated-ETL-Data-Pipeline-BI-Analytics-Platform
```

Tareeqa B (Bina Git ke - Direct ZIP Download):

1. Browser mein jayein: https://github.com/SamiSiddiqui45/Enterprise-Automated-ETL-Data-Pipeline-BI-Analytics-Platform
2. Green color ke **<> Code** button par click karke **Download ZIP** karein.
3. ZIP ko Extract karein aur VS Code mein **File** → **Open Folder** karke folder open kar lein.

Step 2: Dependencies Install Karein & Config Banayein

VS Code terminal mein yeh do commands run karein:

```
# 1. Required Python packages install karein:
pip install -r requirements.txt

# 2. Local environment configuration file banayein:
copy .env.example .env
```

Step 3: SQL Server Mein Database & Star Schema Tables Banayein

Dost ke laptop ke SQL Server mein AnalyticsDW aur tables banane ke 2 tareeqay hain:

Tareeqa A (SSMS ke zariye - Sabse Easy):

1. **SSMS 20** kholein aur .\SQLEXPRESS se Connect karein.
2. Project ke sql/ folder se yeh 7 scripts bari bari khol kar **Execute (F5)** dabayein:
   • 01_create_database.sql (AnalyticsDW database banayega)
   • 02_create_schemas.sql (stg, dw, etl schemas banayega)
   • 03_create_staging.sql (stg.OrdersRaw staging table banayega)
   • 04_create_dimensions.sql (DimDate, DimCustomer, DimProduct, DimGeography, DimShipMode)
   • 05_create_fact.sql (dw.FactSales fact table foreign keys ke saath banayega)
   • 06_create_audit.sql (PipelineRuns aur RejectedRecords audit tables banayega)
   • 07_create_indexes_constraints.sql (Performance indexes banayega)

Tareeqa B (VS Code Terminal se direct commands):

```
sqlcmd -S .\SQLEXPRESS -E -i "sql\01_create_database.sql"
sqlcmd -S .\SQLEXPRESS -E -i "sql\02_create_schemas.sql"
sqlcmd -S .\SQLEXPRESS -E -i "sql\03_create_staging.sql"
sqlcmd -S .\SQLEXPRESS -E -i "sql\04_create_dimensions.sql"
sqlcmd -S .\SQLEXPRESS -E -i "sql\05_create_fact.sql"
sqlcmd -S .\SQLEXPRESS -E -i "sql\06_create_audit.sql"
sqlcmd -S .\SQLEXPRESS -E -i "sql\07_create_indexes_constraints.sql"
```

Step 4: Python Automated ETL Pipeline Run Karein (Data Load)

Abhi database khali hai. Data process aur warehouse mein load karne ke liye VS Code terminal mein run karein:

```
python -m src.main
```

Yeh command kya karegi? Yeh source CSV se 9,994 rows process karegi, staging aur 5 dimension tables bharegi, 9,994 fact records load karegi, aur screen par **Financial Reconciliation PASSED (Zero Variance: $2,297,200.86 Sales & $286,397.02 Profit)** ka report degi!

Step 5: Interactive Live BI Analytics Dashboard Launch Karein

VS Code terminal mein bas yeh likhein:

```
streamlit run src/dashboard.py
```

Browser mein **http://localhost:8501** open ho jayega jisme Executive Overview, Product & Sales Analysis, aur Customer Analytics ke teeno interactive pages live chalenge!

Troubleshooting (Agar Dost ke PC par koi issue aaye)

• **Error: Cannot connect to SQL Server:** Windows Search mein services.msc kholein aur check karein ke **SQL Server (SQLEXPRESS)** Running hai ya nahi.
• **Error: Server Name mismatch:** Agar dost ke PC par instance name .\SQLEXPRESS nahi balki localhost hai, toh bas .env file khol kar DB_SERVER=localhost kar dein.
• **Power BI Desktop Integration:** Agar instructor ne .pbix file mangi ho toh project ke docs/power_bi_guide.md mein tamam connection steps aur DAX formulas ready hain.